MERIT USER GUIDE

# Your daily, weekly and quarterly work

A step-by-step guide to the work most MERIT users do: loading blood test results, clearing held records, keeping laboratory and provider identifiers current, entering people and results by hand, following up cases, and sending the federal extracts.

### In one minute

- **Every day:** upload each laboratory file on **Intake**, then clear **Held records** and **Match review**. Work your tasks in **Work queues**.
- **When a fax or paper report arrives:** search for the person first, then **Enter a result** on their page.
- **Every week:** check overdue tasks, open cases, clusters and the reporter timeliness report.
- **Every quarter:** generate the CDC CBLS file on **Extracts**, review its findings, send it, and mark it submitted. ABLES (NIOSH, adults) is once a year.

Screens shown are from the MERIT demonstration site, with made-up people. What you see depends on your role and profile: a button that is missing for you is a permission you do not have (see the companion guide *Who sees what, and your profiles*).

Companion guides: *MERIT User Guide - Importing NPI and CLIA Records*; *MERIT User Guide - Access and Profiles*. Interactive diagrams: `docs/daily-intake_workflow.html`, `docs/npi-clia-import_workflow.html`, `docs/cbls-extract_lifecycle.html`.

# Contents

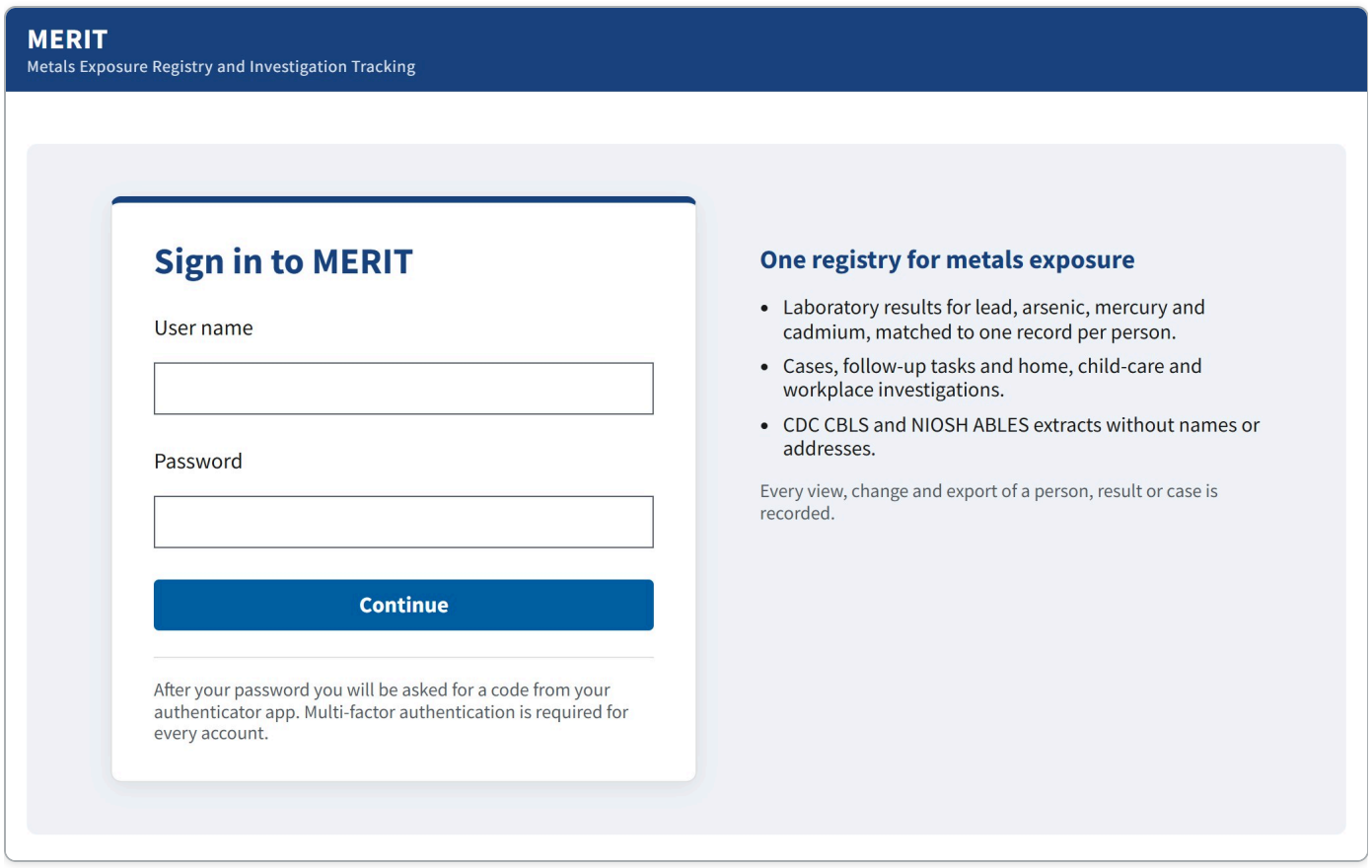

1. Signing in and finding your way
2. Your routine at a glance
3. Importing blood test results
4. Clearing held records
5. Match review: is this the same person?
6. Laboratories, providers, NPI and CLIA
7. Entering people and results by hand
8. Places, relationships, jobs and sources
9. Cases and work queues
10. Investigations
11. Searching, results, clusters, map and reports
12. Federal extracts: CDC CBLS and NIOSH ABLES
13. For supervisors and administrators
14. Who can do what
15. Common questions


# 2. Your routine at a glance

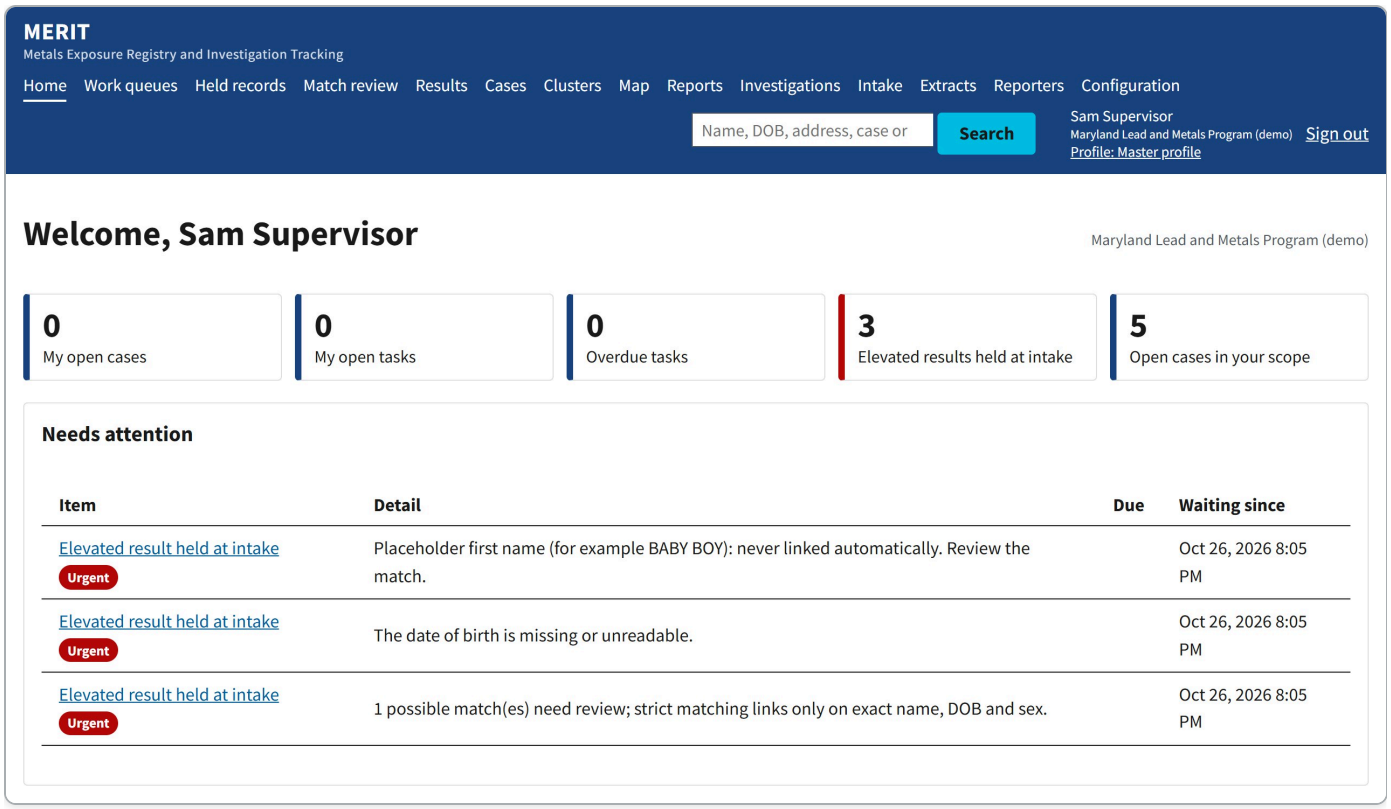
| How often | Task | Where | Usually done by |
|---|---|---|---|
| **Every day** | Upload each laboratory file as it arrives; check the batch card | Intake | Intake clerk, supervisor |
| | Clear held records, urgent (high results) first | Held records | Intake clerk, supervisor |
| | Decide possible person matches | Match review | Intake clerk, supervisor |
| | Enter fax and paper reports | Search, then the person page | Intake clerk, case manager |
| | Work your tasks: confirmatory tests, address review, contacts, notifications | Work queues, Cases | Case manager, investigator |
| **Every week** | Look for overdue tasks and unassigned cases; reassign | Home, Work queues, Cases | Supervisor |
| | Review clusters under review and new cluster alerts | Clusters, Map | Supervisor, epidemiologist |
| | Add new laboratories, facilities and providers; fix "unknown sender" holds | Reporters | Intake clerk, supervisor |
| | Schedule and close investigations | Investigations | Investigator |
| **Every month** | Check reporter timeliness and completeness; case timeliness | Reports | Supervisor, epidemiologist |
| **Every quarter** | Generate, review, send and mark submitted the CDC CBLS file for the quarter that ended | Extracts | Supervisor (epidemiologists can generate) |
| | Review testing counts and elevated counts; export for partners | Reports | Epidemiologist |
| **Every year** | Generate and send the NIOSH ABLES file; review thresholds and settings | Extracts, Configuration | Supervisor, administrator |
| **As needed** | Add users and give access; add a county, program or other tenant | Users, Tenants | Administrator |

# 1. Signing in and finding your way

1. Go to your MERIT address and enter your **user name** and **password**.
2. Enter the 6-digit **code from your authenticator app**. The first time, MERIT shows a QR code and a key: add it to an authenticator app on your phone, type the code it shows, then **save your recovery codes** somewhere safe. Each recovery code works once if you lose your phone.
3. MERIT opens on **Home**. After 20 minutes without activity (your agency may set a different time) you are signed out; unsaved typing is lost, so save as you go.




Sign-in: your user name and password, then **Continue**. The next page asks for the code from your authenticator app.




Home: your counts across the top, **Needs attention** (urgent items first), then a tile for each work queue. Click any count or tile to open the list behind it.

## The header

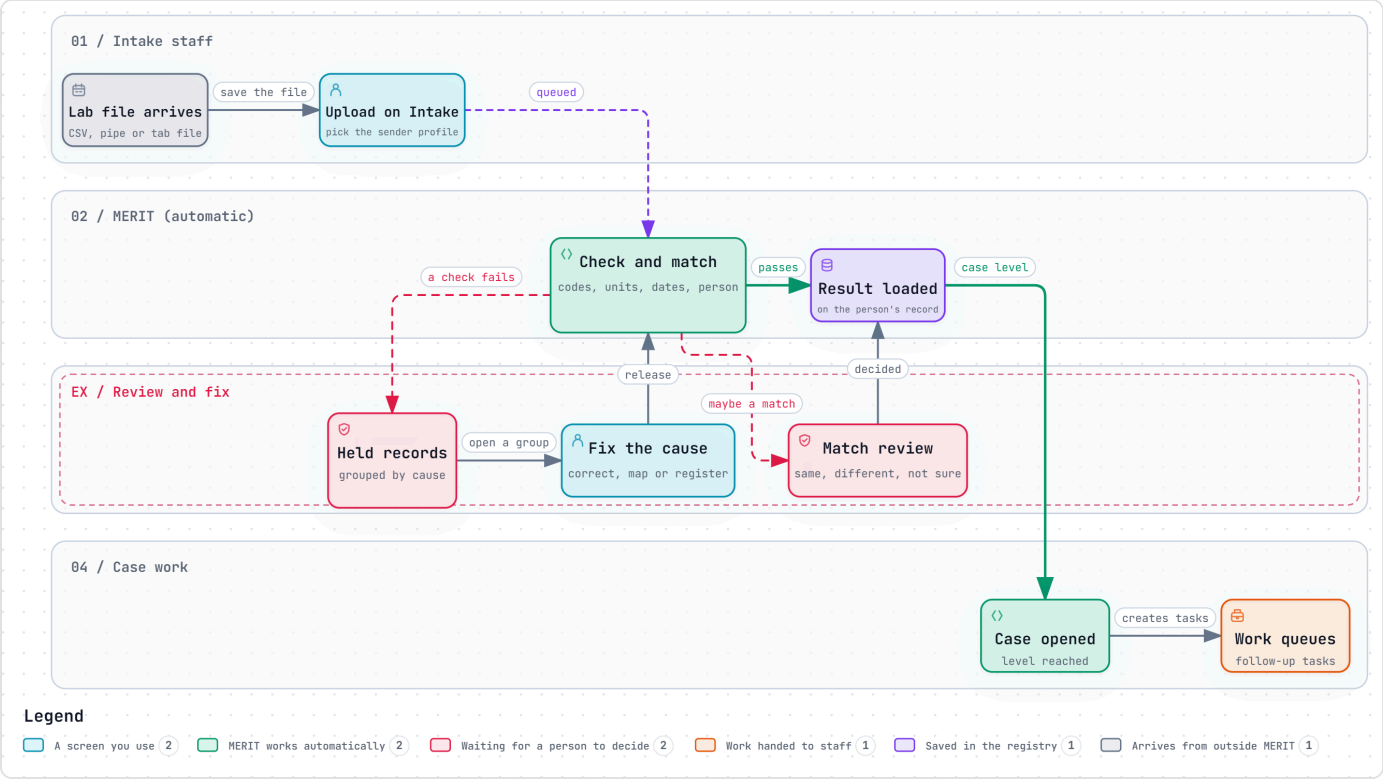
- **Menu:** only the screens your role allows are shown.
- **Search:** type a name, date of birth, address, case number or accession number and press `Enter`.
- **Your name, the agency you are working in, and your profile.** Click **Profile** to switch to a smaller profile (for example read-only) or back to your master profile.

> **Tip:** every screen works with the keyboard alone. `Tab` moves between fields, `Enter` presses the focused button, and the first `Tab` on a page offers *Skip to main content*.

# 3. Importing blood test results

Who: anyone with the *Upload intake files* permission (intake clerks and supervisors).

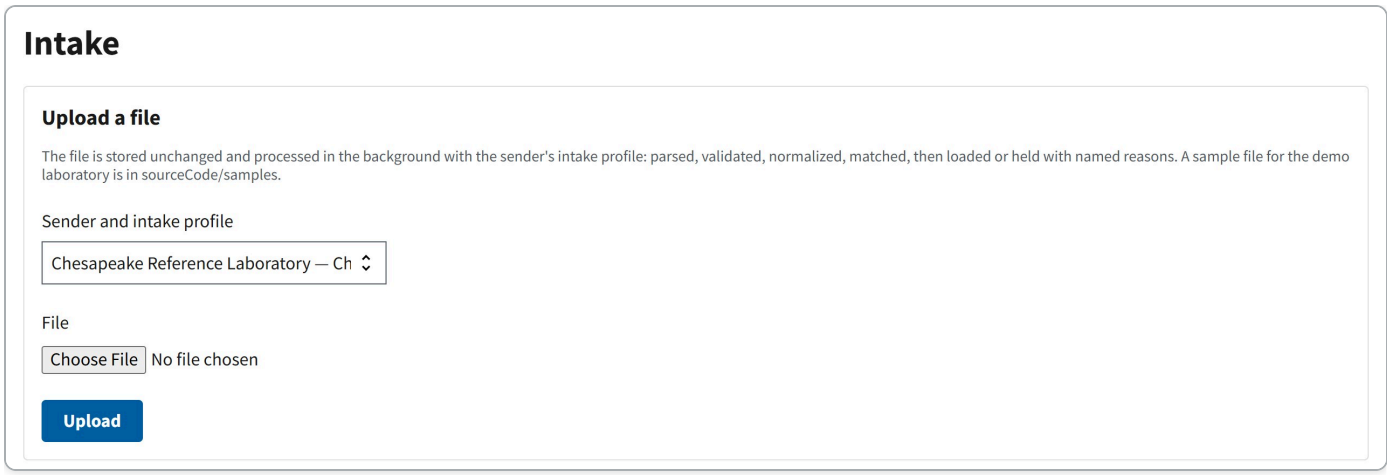
Laboratories send results as files (CSV, pipe- or tab-delimited). Each laboratory has an **intake profile** that tells MERIT how to read its file. MERIT stores the file unchanged, then checks every row in the background, usually within a few minutes.




The daily intake workflow. Interactive version: `docs/daily-intake_workflow.html`.

## Upload a file

1. Open **Intake**.
2. Under **Sender and intake profile**, choose the laboratory that sent the file. If the laboratory is not listed, ask your administrator to set up its intake profile.
3. Click **Choose File**, pick the file, then click **Upload**.
4. Watch the batch card under **Recent batches**. It updates by itself until processing ends.




Upload a file: choose the sender, choose the file, then **Upload**.




A batch card under **Recent batches**: how many rows were **received**, **loaded**, **held** and **rejected**. Click **held** to go straight to those rows.

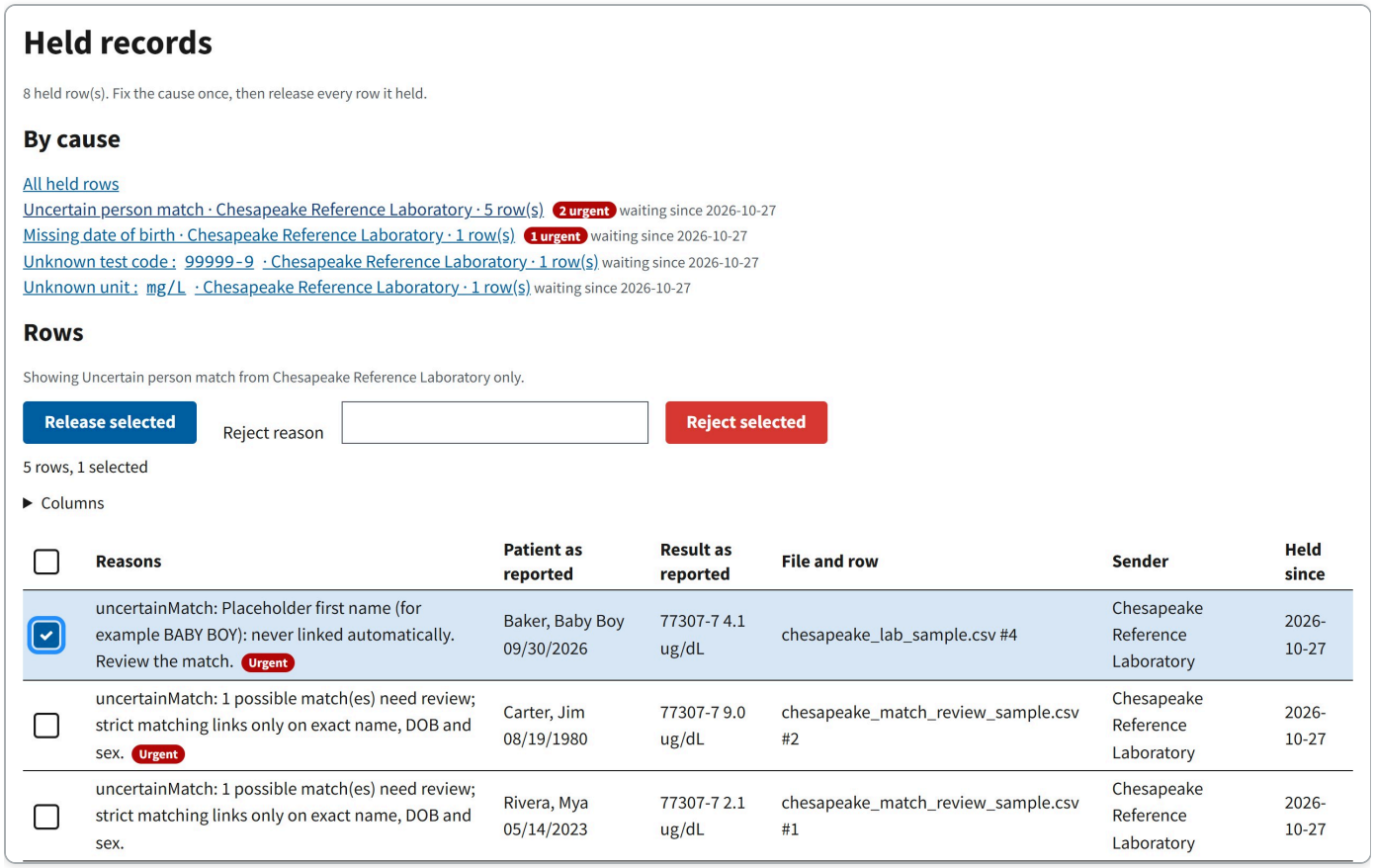
| The card says | What it means | What to do |
|---|---|---|
| Loaded | Every row was checked; good rows are on people's records | Clear any held rows (next section) |
| Processing | MERIT is still working | Wait; the card refreshes itself |
| Failed | The file could not be read (wrong format or sender) | Check you chose the right sender; ask the laboratory for a corrected file |

**The same file twice?** MERIT does not process a file it has already received for the same sender: it says when the file arrived, links to that batch, and loads nothing. If the laboratory really sent the same file again, choose it again and tick **This is a genuine re-send**. Corrected results should come from the laboratory marked as corrected.

# 4. Clearing held records

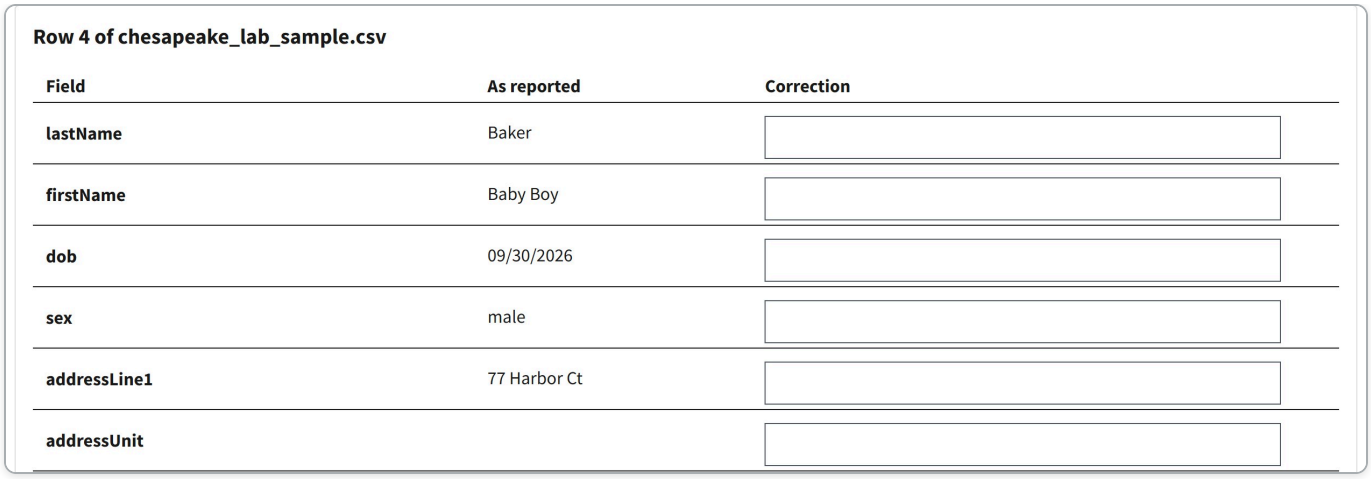
Who: anyone with the *Release held results* permission (intake clerks and supervisors).

A row is **held** when MERIT cannot load it safely. It waits with a named **cause**. Held records are grouped by cause, laboratory and reported value, so you fix a cause once and release every row it held. Groups with high results are marked **urgent**: work them first.




Held records: groups by cause at the top, rows below. Selecting a single row opens its reported values beside a correction form.

1. Open **Held records** (or click **Held intake records** on Home).
2. Click a group under **By cause**. Its rows appear under **Rows**.
3. Fix the cause (table below). To correct one row, tick only that row: type the right value in the **Correction** column and click **Save correction**. The reported value is never changed; the correction sits beside it. You can correct one field, save, then correct another: every saved correction is kept. To remove a correction, clear its box and save.
4. Tick the rows and click **Release selected**. MERIT checks them again; rows that now pass are loaded.
5. If a row should never load (a test row, or a result for another state), type a **Reject reason** and click **Reject selected**.

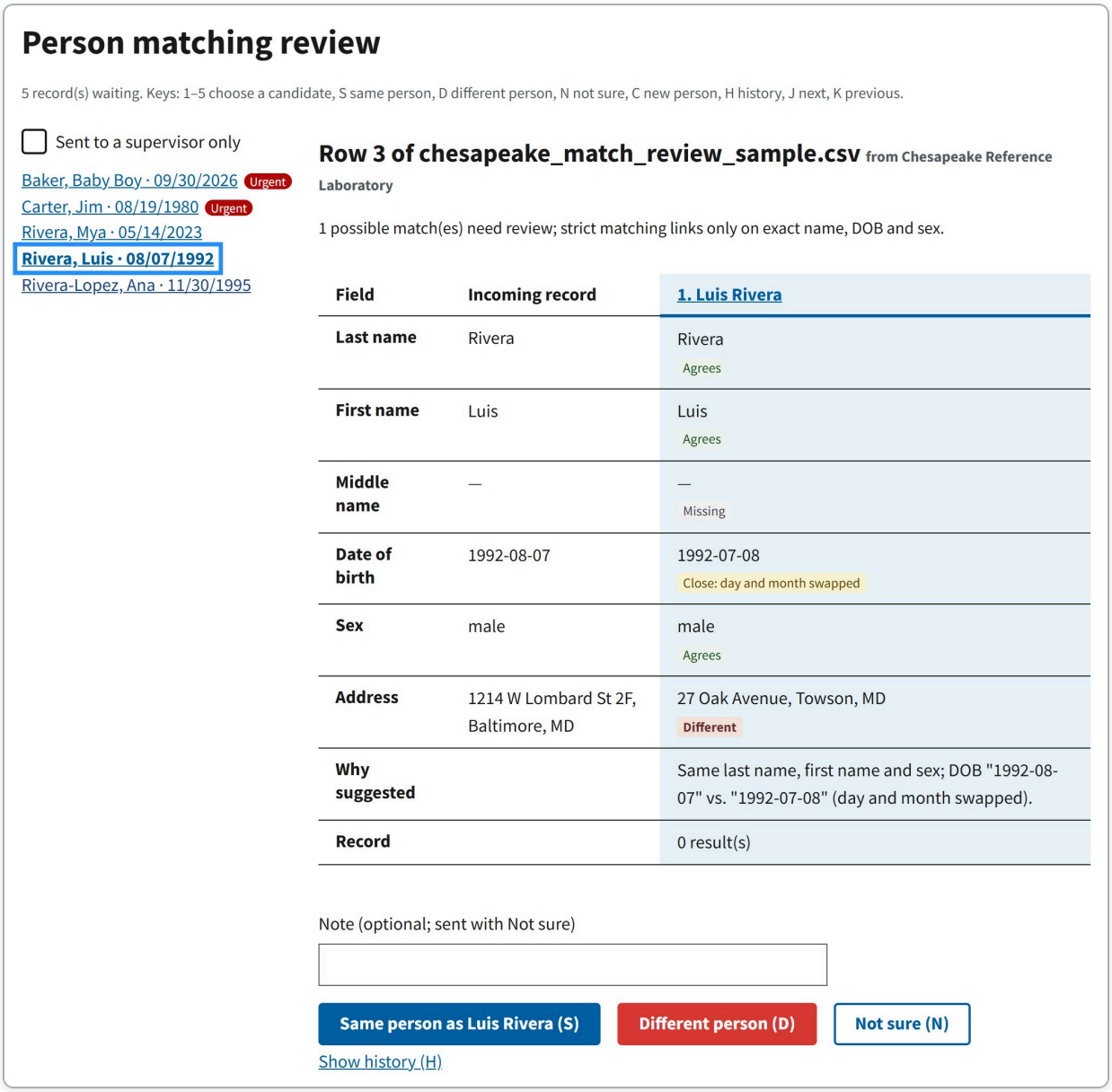



Correcting one row: the reported values stay as sent; type the right value in **Correction** and click **Save correction** below the list.

| Cause | Usual fix |
|---|---|
| Unknown sender | The laboratory's CLIA number is not in the registry. Add it on **Reporters** (section 6), then release. |
| Unknown test code | The LOINC code is not mapped to a metal. Ask your administrator to map it, or correct an obvious typo, then release. |
| Unknown unit | The unit is not accepted for this test (for example mg/L for blood lead). Confirm the right unit with the laboratory and correct it. |
| Missing date of birth / missing result value / impossible dates | Call the laboratory or the ordering provider; correct the field. |
| Uncertain person match | Decide it on **Match review** (next section). |
| Possible duplicate | Compare with the result already loaded; reject the duplicate with a reason. |
| Out of jurisdiction / unusable address | Check the address; correct it, or reject with a reason and forward the report as your program requires. |

# 5. Match review: is this the same person?

MERIT links a result to a person automatically only when name, date of birth and sex agree exactly. When a record looks like someone already in MERIT but not exactly (a nickname, a swapped day and month, a hyphenated name, a possible twin), it waits for you on **Match review**. Placeholder names such as "Baby Boy" are never linked automatically.



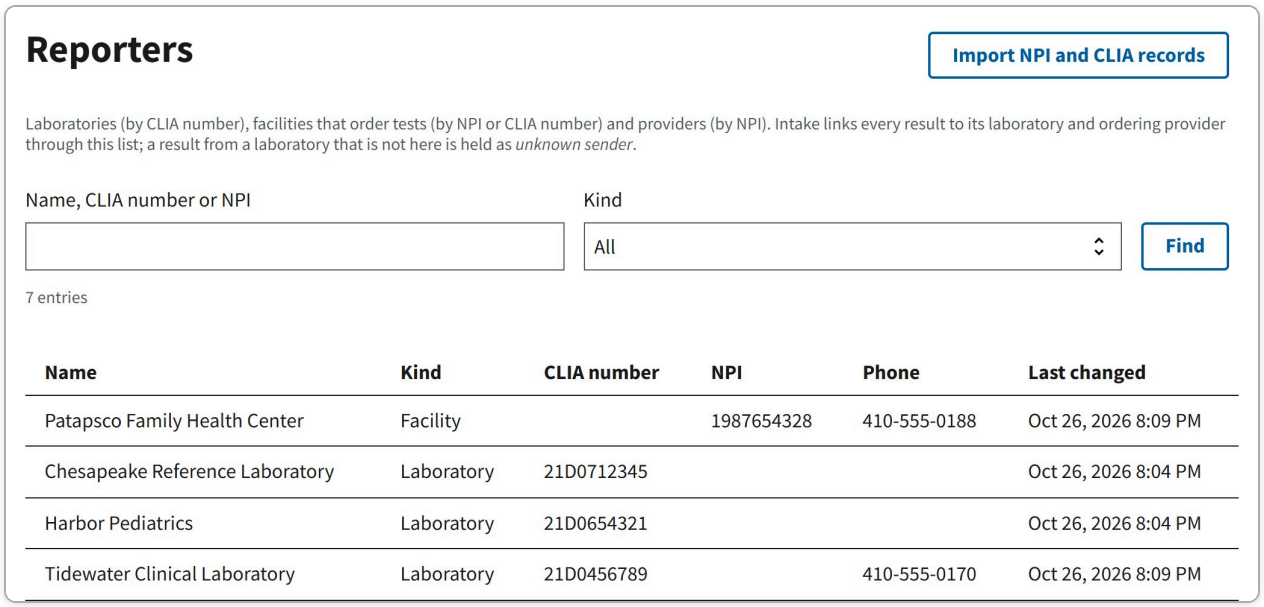

Match review: the waiting records on the left (urgent first); the incoming record and each candidate side by side, field by field, with **Why suggested** and the decision buttons below.

| Decision | Key | What happens |
|---|---|---|
| Same person as … | S (or 1–5 to pick a candidate) | The result is linked to that person |
| Different person | D | MERIT remembers the pair and will not suggest it again; with no candidate left, the record loads as a new person |
| Not sure | N | Sent to a supervisor, with your optional note |
| Create as a new person | C | A new person record is created |
| History / next / previous | H J K | Show the candidate's history; move through the queue |

> **When in doubt, choose Not sure.** Linking two different people is much harder to undo than waiting a day for a supervisor. Supervisors tick **Sent to a supervisor only** to see the records escalated to them.

## 6. Laboratories, providers, NPI and CLIA

The **Reporters** screen is MERIT's list of who reports results: **laboratories** (identified by their CLIA number), **facilities** that order tests such as clinics and hospitals (by NPI or CLIA number), and individual **providers** (by NPI). Intake uses it to link each result to its laboratory and ordering provider. A result from a laboratory that is not on the list is held as **unknown sender**.




Reporters: search by name, CLIA number or NPI; filter by kind. People with *Edit the reporter registry* also see the Add form and **Import NPI and CLIA records**.

### Add or correct one entry

1. Open **Reporters** and search first: the entry may already be there.
2. Under **Add or correct one entry**, choose the **Kind**. The form shows that kind's fields: the name and CLIA number for a laboratory or facility, the last name, first name and credential for a provider, and the NPI and phone for all.

3. Click **Save entry**. An entry with the same CLIA number or NPI is **updated**, never duplicated. A laboratory and a facility with the same CLIA number or NPI are one organization.
4. If rows were held as *unknown sender*, go to **Held records** and release them.

> **Many entries at once?** Use **Import NPI and CLIA records** with a CSV file. The separate guide *MERIT User Guide - Importing NPI and CLIA Records* walks through it.

# 7. Entering people and results by hand

For fax and paper reports, phone calls, and records from other programs. Who: anyone with *Edit people* and *Enter results*.

## Always search first

Use the header search with the person's name and date of birth. MERIT keeps **one record per person**; a duplicate splits their history and can hide a case. Open the person if they are already there.

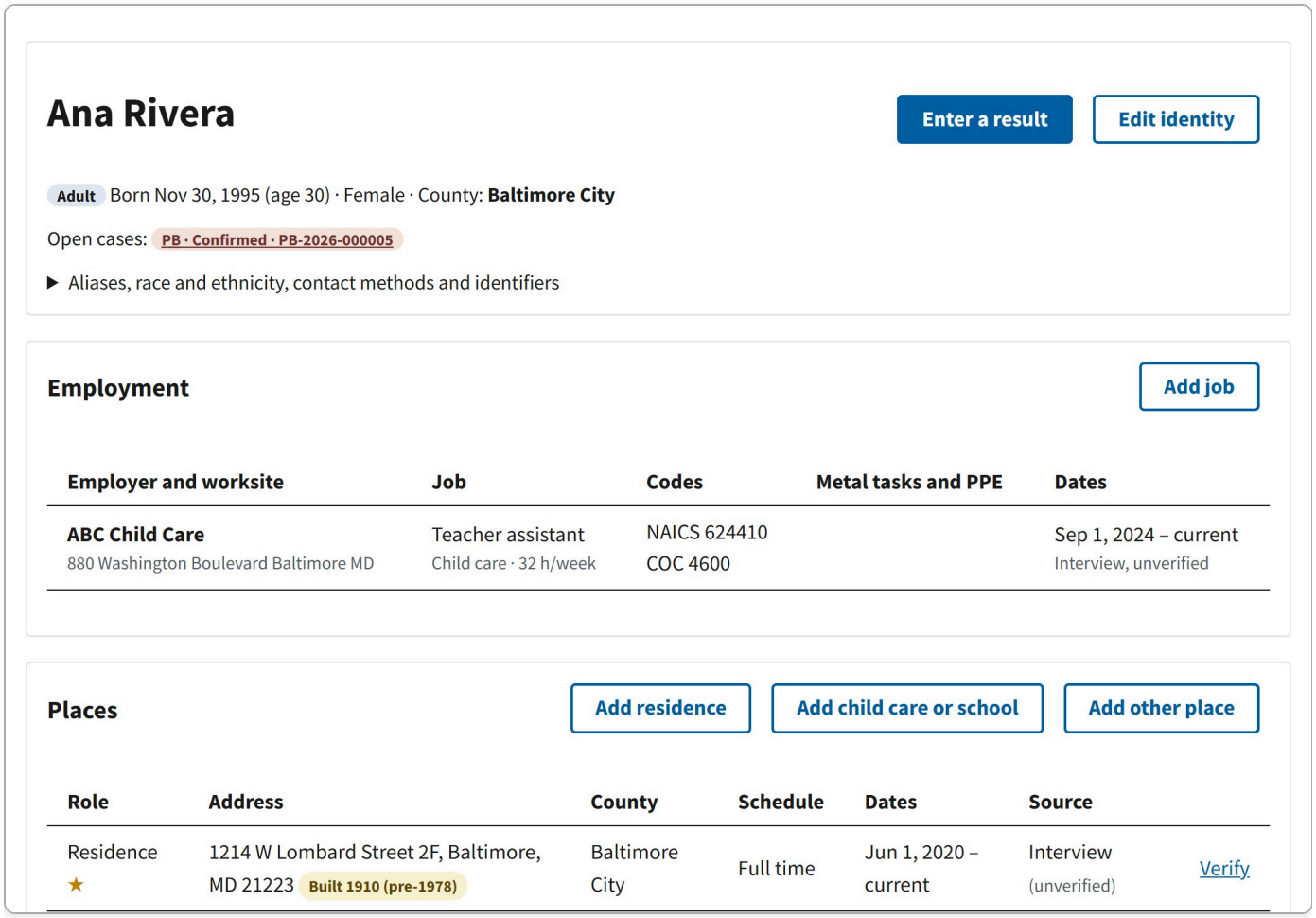
## Add a person

1. From the search results, choose **Add a person**.
2. Enter last, first and middle name, date of birth (tick **Verified against a second source** when you have checked it), sex, race (any that apply), ethnicity and preferred language.
3. Save. The person page opens. A person record holds identity only; places, relationships and jobs are added on the person page.




**Add a person.** Last and first name, date of birth (tick **Verified against a second source** once checked), sex, race, ethnicity and preferred language, then **Save**.

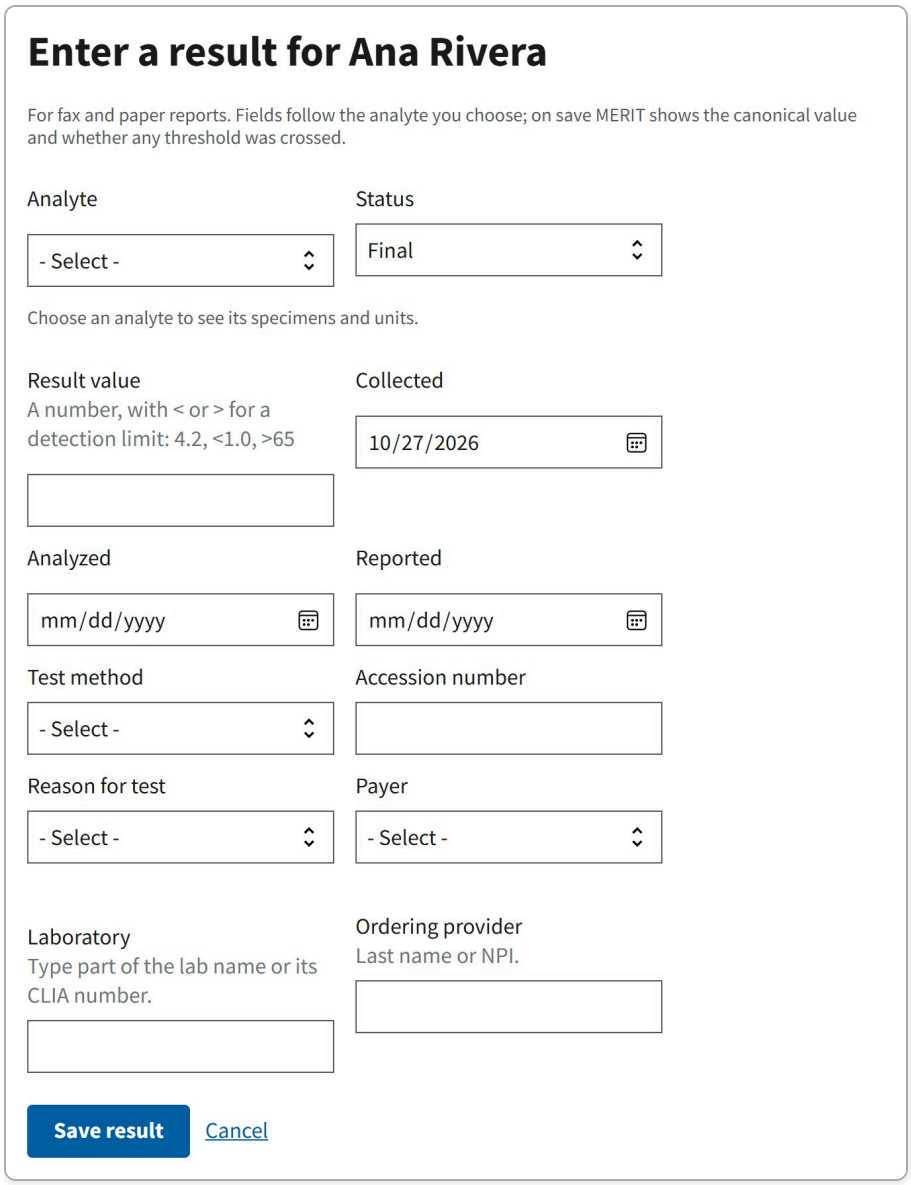
## The person page




The person page: identity at the top (**Enter a result**, **Edit identity**), then a section for each kind of information, in the order the person's population needs them.

## Enter a result (fax or paper)

1. On the person page click **Enter a result**.
2. Choose the **Analyte** (lead, arsenic and so on). The specimen and unit choices then follow the analyte.
3. Enter the **Result value** exactly as reported. Use `<` or `>` for a detection limit: `4.2`, `<1.0`, `>65`.
4. Enter the **Collected** date (required), and the analyzed and reported dates, test method, accession number, reason for test and payer when you have them.
5. Save. MERIT shows the standardized value and whether it reached any level. A result at or above the case level opens or updates a case, exactly as for an uploaded file.




**Enter a result.** Choose the analyte first: the specimen and unit choices follow it. Type the value as reported, the collected date, and the other details you have, then **Save result**.

## 8. Places, relationships, jobs and sources

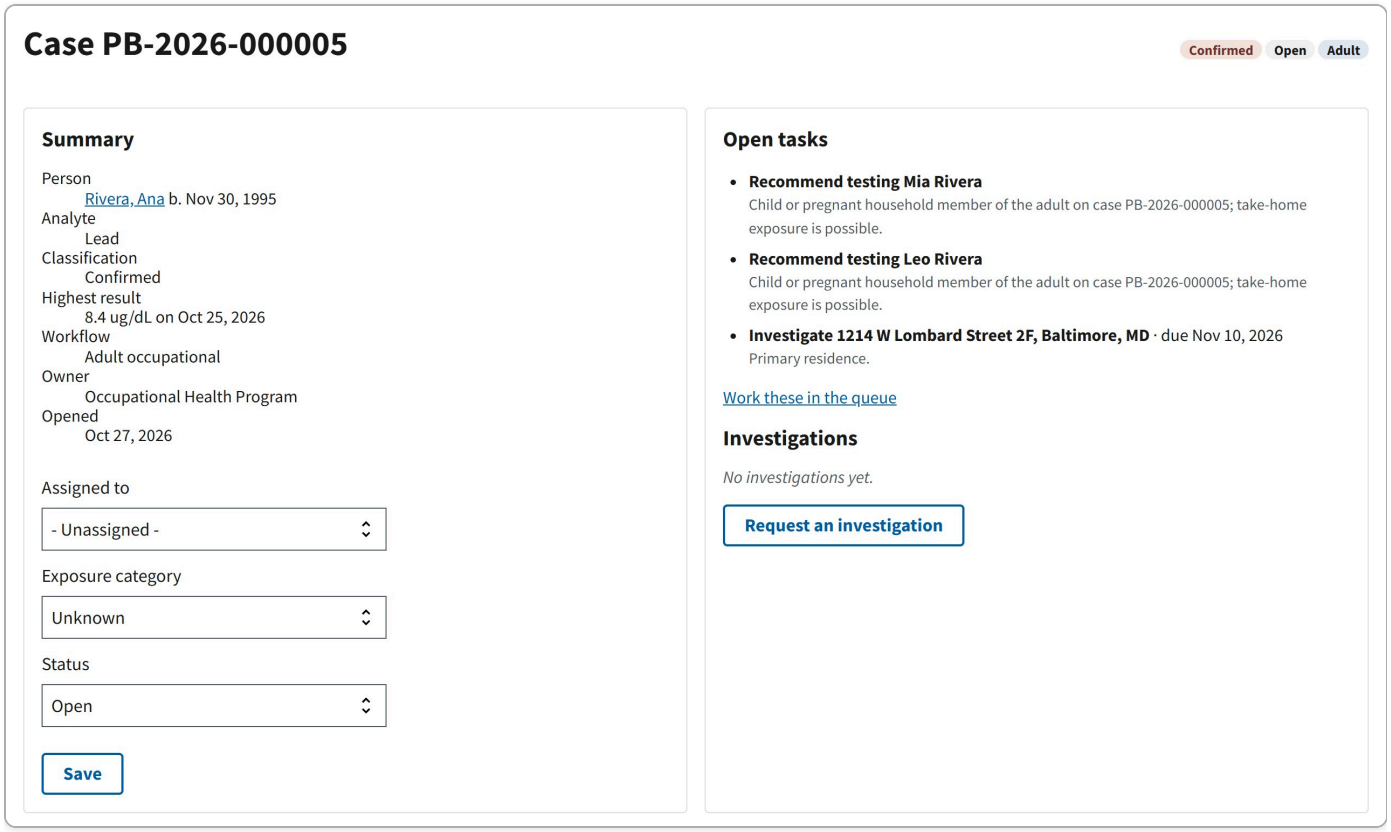
Each is a section of the person page with its own **Add** button. Every entry records its **source** (interview, laboratory report, and so on) and dates, so history is kept rather than overwritten.

| Section | Buttons | Good to know |
|---|---|---|
| **Places** | Add residence · Add child care or school · Add other place · Set as primary · Verify · End | Enter the street address, city, state and ZIP; MERIT finds the county. For a new residence choose whether it **becomes primary from its start date** or the current primary **stays primary**; overlapping residences need a reason. Add the year built for housing-age risk. For child care and other places, give the schedule (hours per week) so MERIT can tell when a site investigation is needed. |
| **Relationships** | Add relationship · End | Search for the other person or enter their name and date of birth, then choose what they are to this person (parent, guardian, sibling, household member…). Mark the **primary contact** and who **has legal custody**. Household children and pregnant members of an adult case get testing recommendations automatically. |
| **Employment** | Add job | Employer and worksite, job title, industry and occupation codes, hours, metal tasks and protective equipment. Needed for adult cases and the ABLES file. |
| **Exposure sources** | Add source | Source type, since when, and a description (product, brand, country of origin). Tick **Confirmed by a sample** after an environmental sample confirms it. |
| **Pregnancies** | Add episode | Start date and expected end. While an episode is open, results count in the pregnant population and prenatal notifications are created. |
| **Timeline** | Open the interactive timeline | Every result, case, place and job on one time line. |

> **Change, do not retype.** To correct a name or date of birth use **Edit identity**; to record a move add a new residence with its start date. Never create a second person to "start fresh".

# 9. Cases and work queues

MERIT opens a **case** automatically when a result reaches the case level for the person's population (child, adult, pregnant). The case collects the results, tasks, activity and investigations for that exposure.

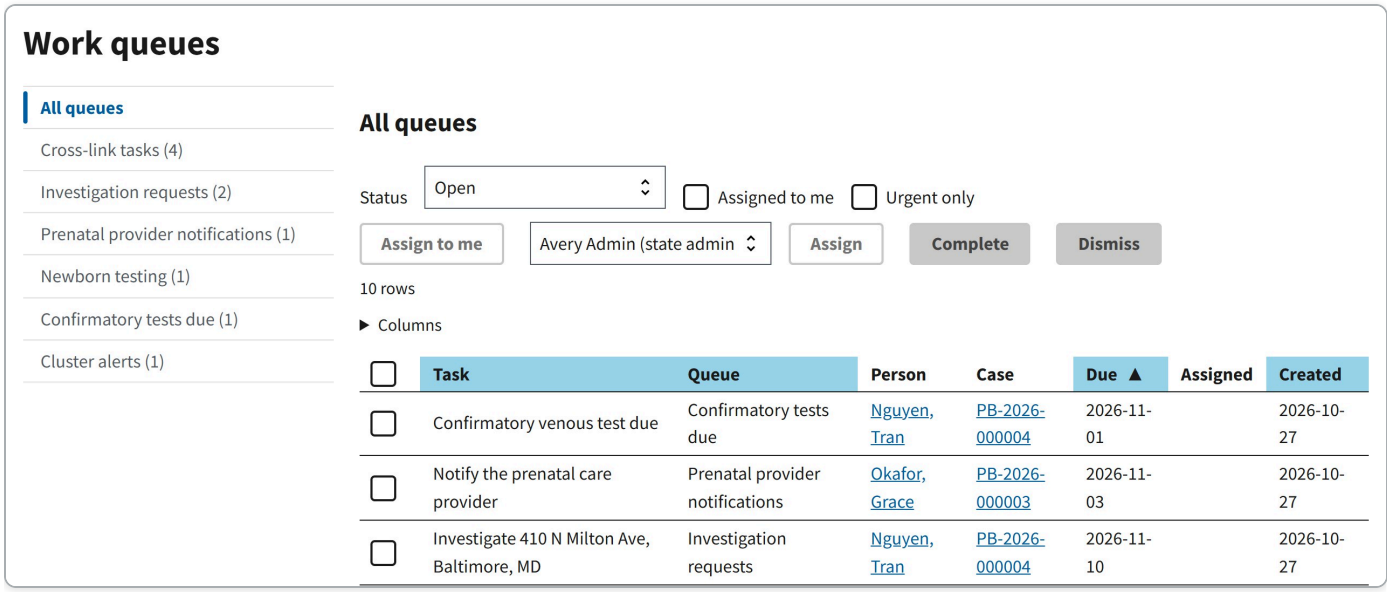



A case: the summary, assignment and status on the left; open tasks and investigations on the right.

## Working a case

1. Open it from **Cases**, from Home, from a task, or from the person page.
2. Set **Assigned to**, the **Exposure category** and the **Status**, then save.
3. Record every call, letter and visit in the **Activity log**. Contact attempts count toward the letter MERIT queues after repeated failed attempts.
4. Click **Request an investigation** for a home, child-care or workplace investigation (section 10).
5. If the automatic classification is wrong, use **Override classification** and give the reason; the override and its reason stay on the case. When follow-up is complete, **Close case** with the closing reason.

## Work queues



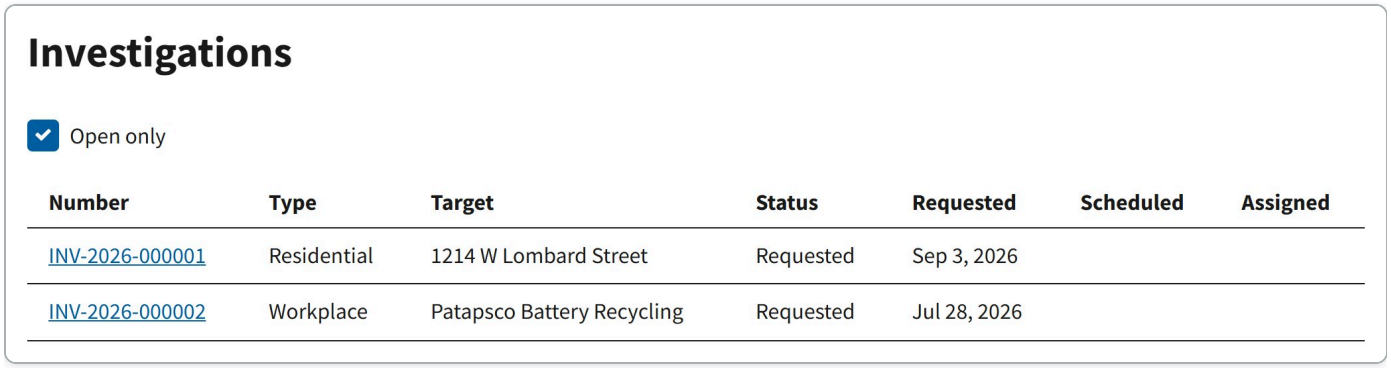

Work queues: pick a queue on the left; filter by status, **Assigned to me** or **Urgent only**; tick tasks and use the buttons above the table.

| Queue | What it asks you to do |
|---|---|
| Confirmatory tests due | A capillary result needs a venous test: contact the family or provider |
| Address review | A laboratory report gave a new address: decide whether the person moved |
| Investigation requests | Schedule the requested home, child-care or workplace investigation |
| Prenatal provider notifications / Newborn testing | Notify the prenatal care provider; make sure the newborn is tested |
| Cross-link tasks | Recommend testing for household members of a case |
| Cluster alerts | Several cases share a building, child-care site or worksite: review the cluster |
| Person match review (supervisor) | A reviewer was not sure; decide on Match review |

Tick one or more tasks, then **Assign to me**, choose a person and **Assign**, **Complete**, or **Dismiss** a task that no longer applies. Change **Status** to see done and dismissed tasks.

# 10. Investigations

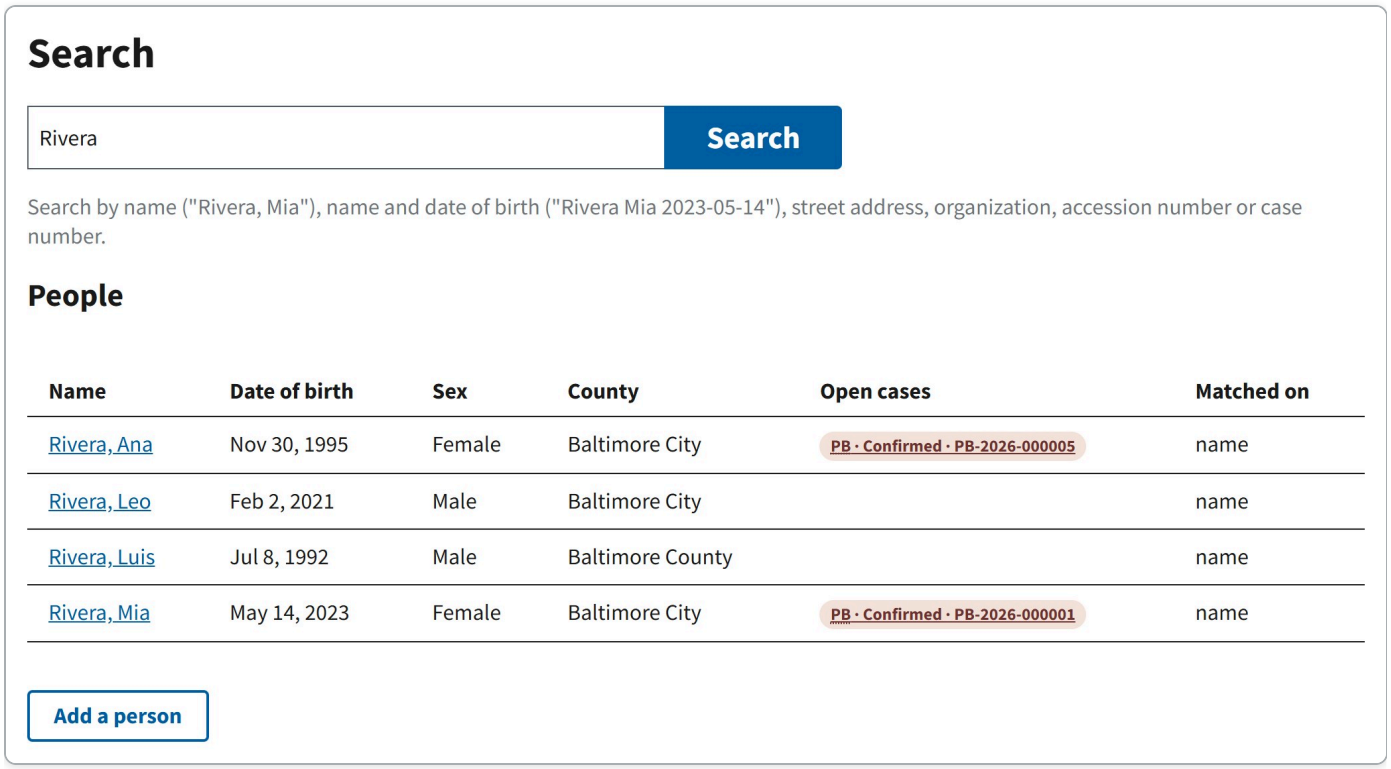
Who: investigators and occupational investigators; supervisors.



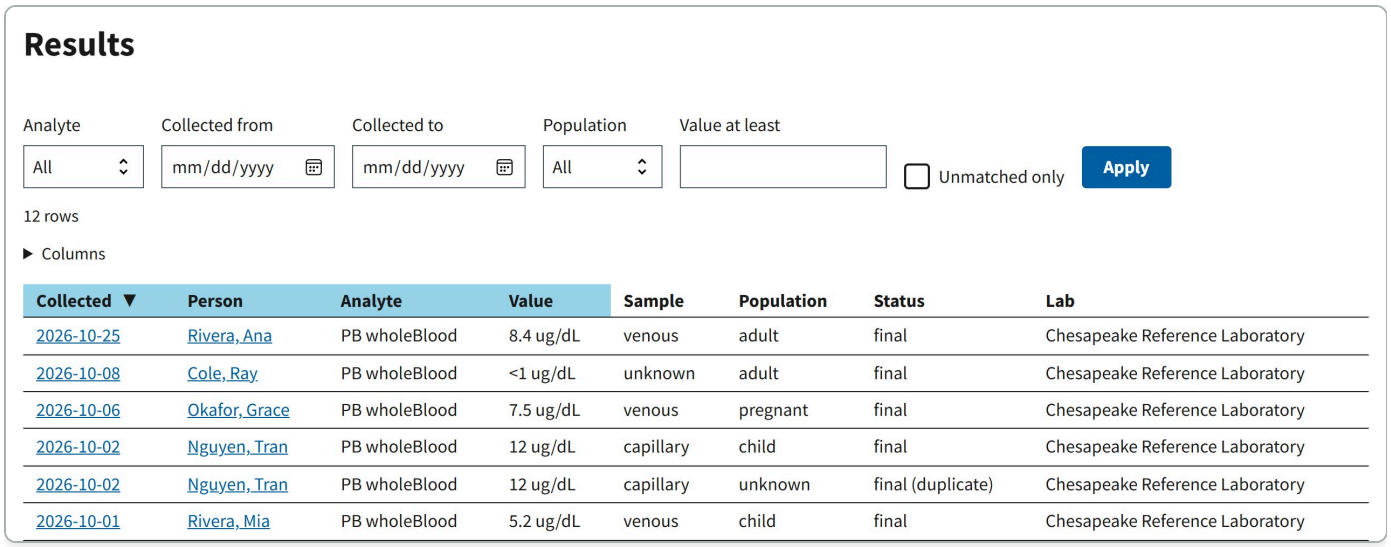

Investigations: number, type (residential, child care, workplace), target, status and dates. Untick **Open only** to see closed ones.

1. A case manager requests an investigation from the case (**Request an investigation**). One investigation of a place can serve several cases; MERIT offers an open one at the same place.
2. Open it from **Investigations**, set the scheduled date and who is assigned, and save.
3. Record **environmental samples** (**Add sample**): where, what (dust, soil, water, paint, product), the result and unit. MERIT compares each with the standard in force.
4. Record the findings and any orders or referrals (for example to OSHA), set the status to complete, and save. The cases served show the outcome.

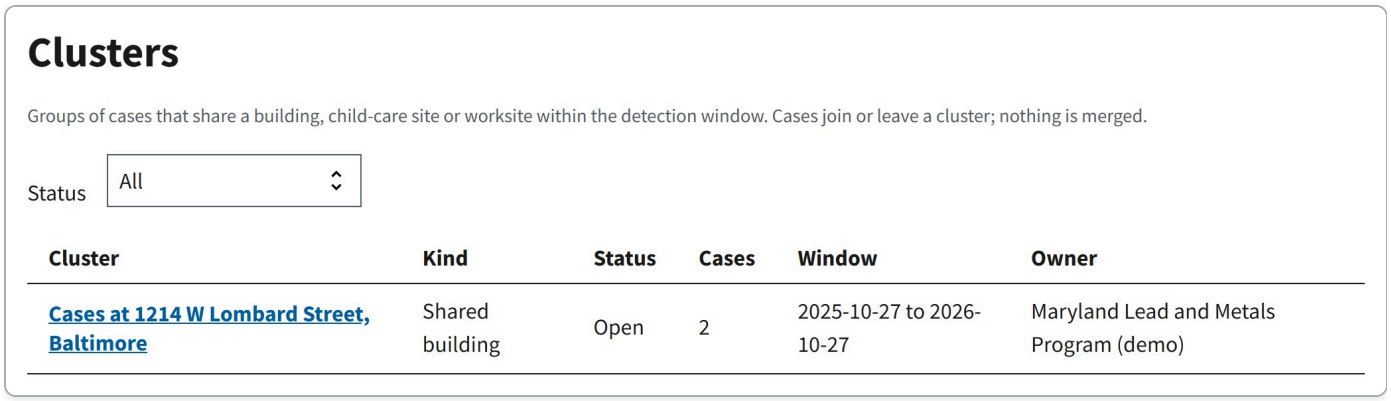
# 11. Searching, results, clusters, map and reports



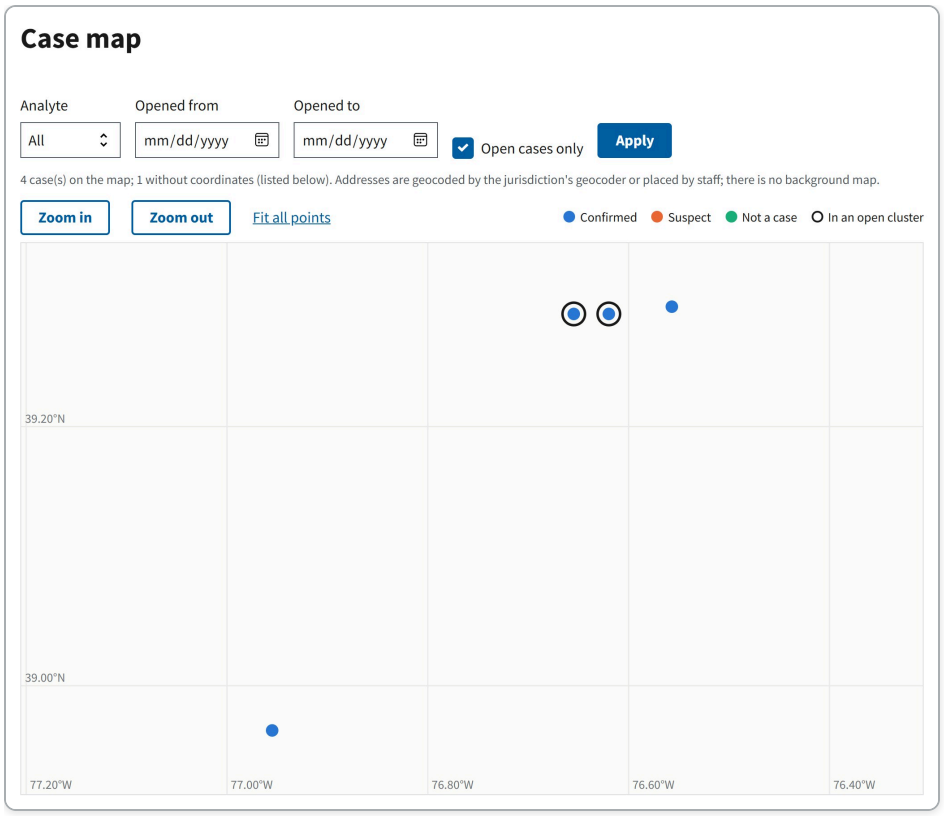

**Search** (header): people, cases and accession numbers. Each open case shows its metal, classification and number, and opens the case. **Add a person** is at the bottom of the results.




**Results**: every result you may see; filter by analyte, collection dates, population and value, and sort by any column.



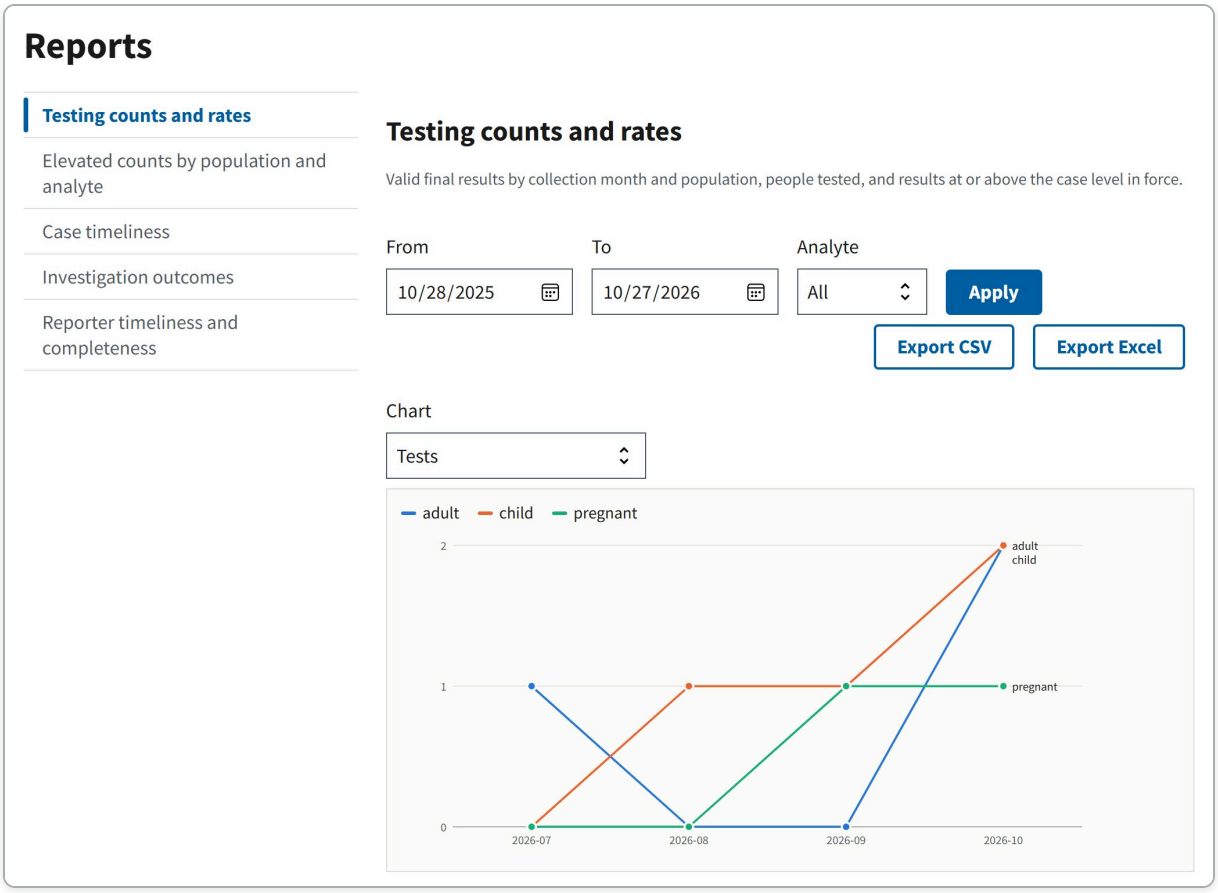

**Clusters**: cases sharing a building, child-care site or worksite within the detection window. Cases join or leave a cluster; nothing is merged.




**Map**: cases by residence, ringed when in an open cluster. Zoom and pan with the buttons, mouse or keyboard; select a point to read it. Cases without coordinates are listed below the map.

## Reports

Choose a report on the left, set **From**, **To** and the analyte, click **Apply**, and use **Export CSV** or **Export Excel** to share it. Exports are recorded in the audit log.



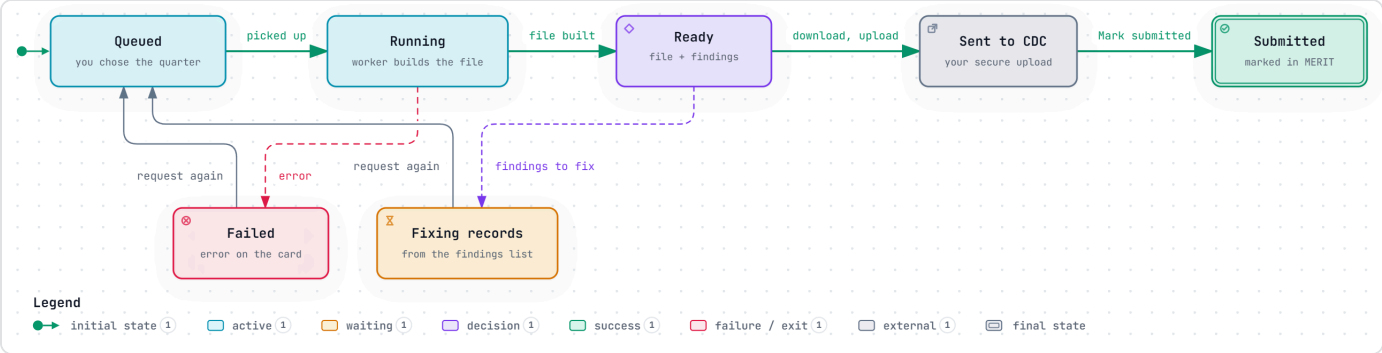

**Testing counts and rates**: tests by month for each population; the table below the chart has the numbers.

| Report | Use it to |
|---|---|
| Testing counts and rates | See tests, people tested and results at or above the case level by month and population |
| Elevated counts by population and analyte | Track elevated results for partners and grant reports |
| Case timeliness | Check how quickly results arrive, cases are opened and families are first followed up (a contact attempt, letter, education, referral or medical management), and how many cases have no follow-up yet |
| Investigation outcomes | Summarize findings, sources and orders |
| Reporter timeliness and completeness | Find laboratories that report late or leave fields blank, and follow up with them |

# 12. Federal extracts: CDC CBLS and NIOSH ABLES

Who: *Generate federal extracts* (supervisors, epidemiologists); *Mark extracts submitted* (supervisors).

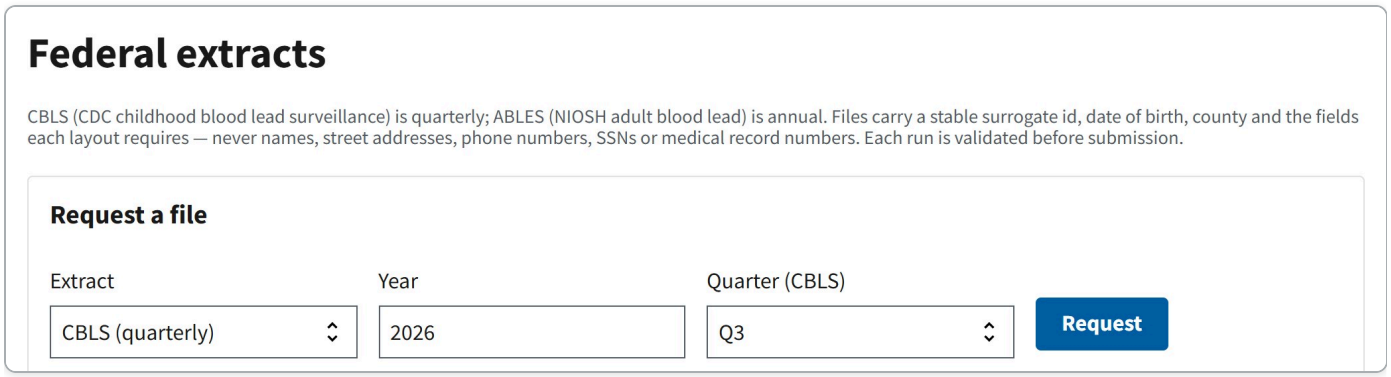
**CBLS** is CDC's Childhood Blood Lead Surveillance file, sent **every quarter**. **ABLES** is NIOSH's Adult Blood Lead Epidemiology and Surveillance file, sent **once a year**. Both are built from the results already in MERIT; you do not re-enter anything. The files carry a stable surrogate id, date of birth, county and the fields each layout needs, never names, street addresses, phone numbers or record numbers.




An extract run from request to submitted. Interactive version: `docs/cbls-extract_lifecycle.html`.

## The quarterly CBLS file, step by step

1. **Before you start** (the week the quarter ends): clear held records and match review, so every result of the quarter is loaded and linked.
2. Open **Extracts**. The form starts on the quarter that just ended. Check **Extract** = CBLS (quarterly), the **Year** and the **Quarter**, then click **Request**.
3. A card appears: **Queued**, then **Running**, then **Ready**. It refreshes by itself.
4. On the ready card, read the record count and the number of **validation findings**. Click **View findings** to see each one: the surrogate record id, the field and the rule.
5. Fix what you can in MERIT (for example a missing county or a wrong date of birth) and click **Request** again for a fresh file. Findings are warnings: your program decides whether a file with findings may be sent.
6. Click **Download** and upload the file to CDC the way your CDC program contact has set up.
7. When CDC has received it, click **Mark submitted** and confirm. The card shows **Submitted**, and the file can still be downloaded later.




Request a file: the extract, the year and, for CBLS, the quarter.




A ready run: the record count and the number of validation findings (**View findings** appears when there are any), **Download**, and **Mark submitted** once the file is sent.

### The annual ABLES file

The same steps, with **Extract** = ABLES (annual) and the year. ABLES uses each adult's employment (industry and occupation codes) and exposure sources, so check the Employment section of adult cases during the year.

> **A run failed?** The card shows the error. Most failures are an inactive layout or missing setting; tell your administrator, then request again.

## 13. For supervisors and administrators

### Weekly checks for supervisors

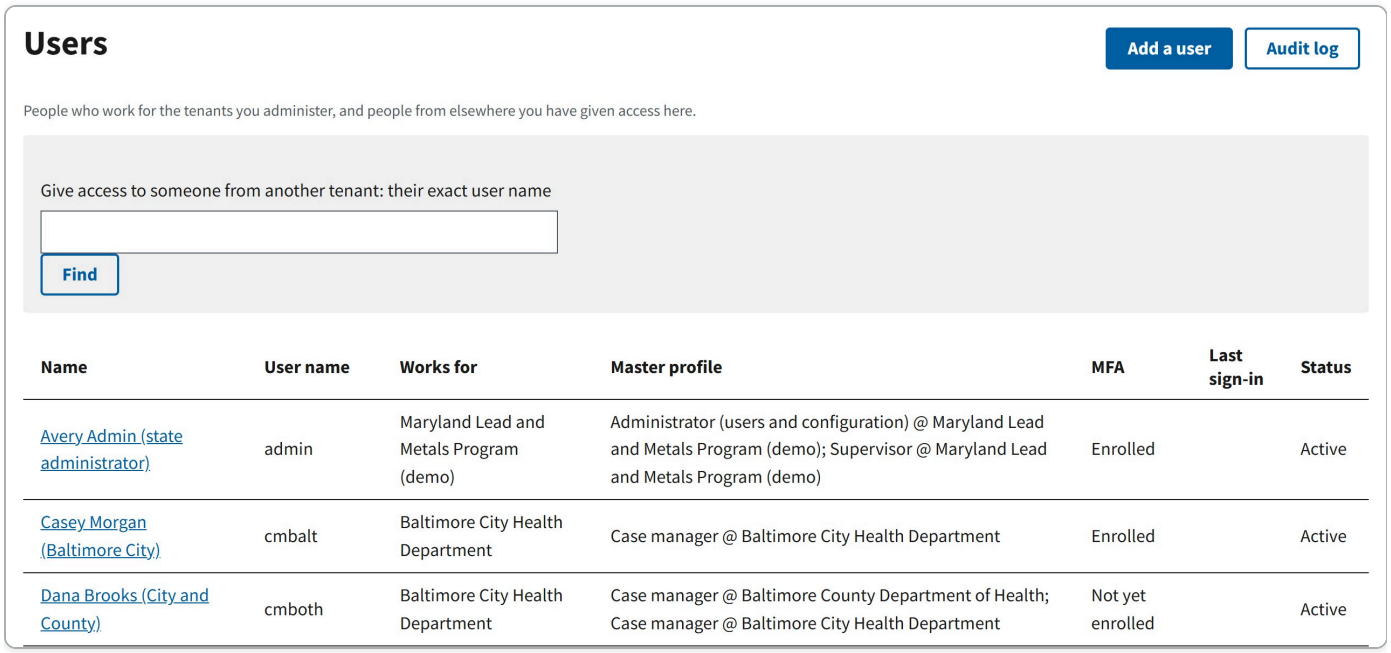
- **Home:** overdue tasks and elevated results held at intake should be near zero.
- **Work queues:** filter **Urgent only**; reassign tasks of people who are away.
- **Match review:** tick **Sent to a supervisor only** and decide the escalated records.
- **Clusters:** review new clusters and set their status.

### Configuration (administrators)

**Configuration** holds the **thresholds** (case and action levels for each metal, specimen and population, with the date each takes effect), the **jurisdiction settings** (for example the investigation look-back, contact attempts before a letter, cluster window and size) and the **analytes** in use. A new value takes effect from its date; old values are kept for history. Review them once a year and whenever guidance changes.

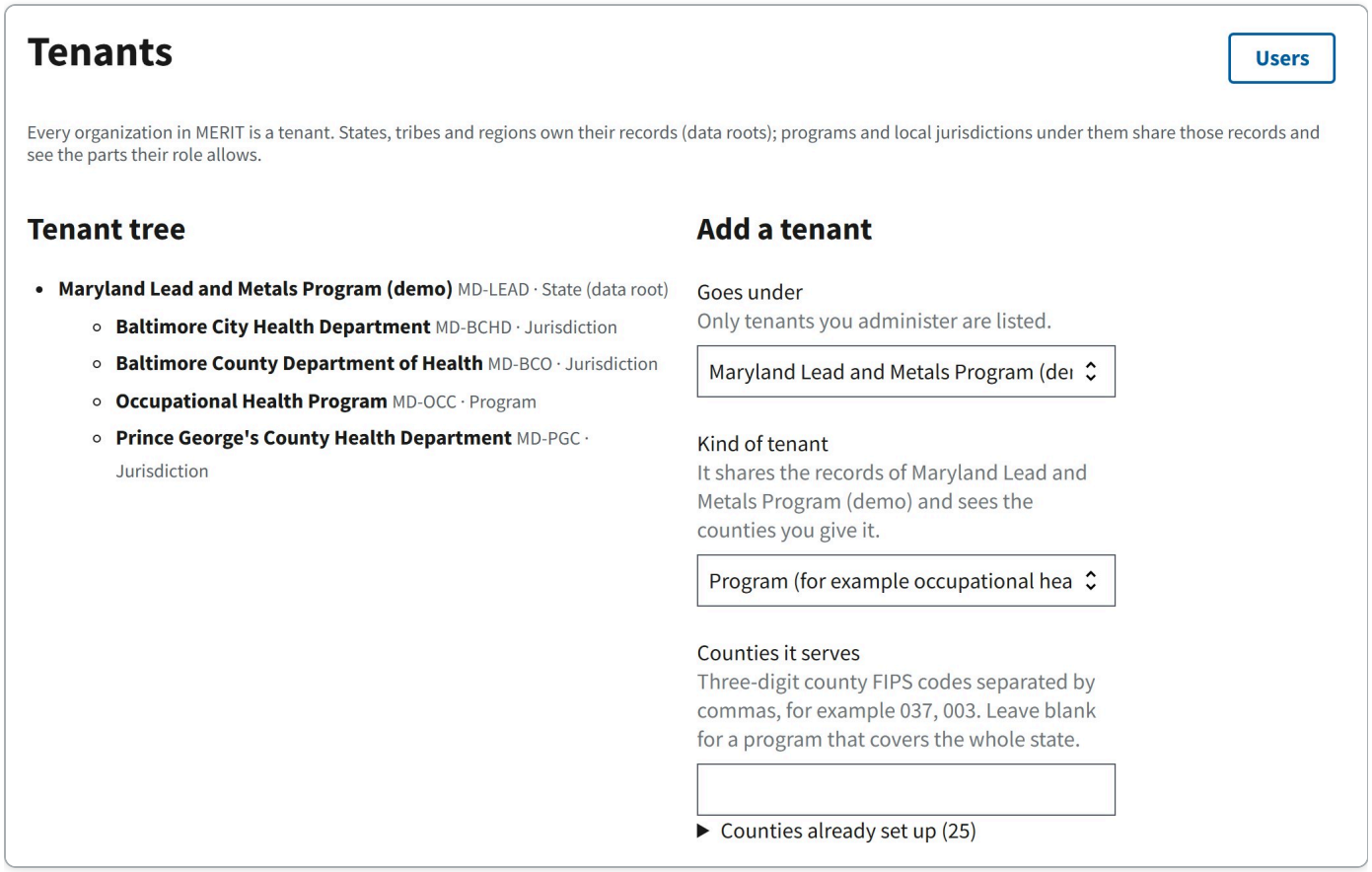
## Users and access

On **Users**, add people, set where they work, and give them roles at a tenant. The companion guide *Who sees what, and your profiles* explains roles, master profiles and viewing as someone.




Users: everyone who works for the tenants you administer, with their master profile and second-factor status. **Add a user** is at the top right.

### Adding a county, program or other tenant




Tenants: the tree of agencies on the left; **Add a tenant** on the right. The kinds offered depend on the tenant you choose under **Goes under**.

1. Open **Tenants**. Under **Goes under** choose the parent (only tenants you administer are listed).
2. Choose the **Kind of tenant**: under a state, a *local jurisdiction* (a county or city health department) or a *program* (for example occupational health). A platform administrator can also add regions, states and tribes; a region administrator can add states and tribes to the region.
3. Give a **Code** (capital letters, digits and hyphens, for example `WV-JEFFERSON`; it cannot be changed) and a **Name**.
4. For a county, list the three-digit county FIPS codes it serves (`037, 003`). Counties not yet in MERIT go in **New counties**, one per line: `037 Jefferson County`. Under a county or program you can choose only counties it already serves, and **New counties** is not offered: new counties are added under the state. For a new state or tribe, give its two-letter postal code and state FIPS code.
5. Click **Add tenant**, then add its people on **Users**.

## 14. Who can do what

The usual roles. Your administrator may give you more than one, and your profile may use only part of what you have.

| Role | Main screens | Typical work |
|---|---|---|
| Intake clerk | Intake, Held records, Match review, Reporters, people and results | Load files, clear holds, enter fax and paper reports, keep the reporter registry current |
| Case manager | Cases, Work queues, people, Reports (export) | Follow up cases, contact families and providers, request investigations |
| Investigator / occupational investigator | Investigations, Cases, people | Home, child-care and workplace investigations, samples, findings |
| Supervisor | Everything above, Extracts, audit log | Escalated matches, overrides, reassignments, federal extracts and their submission |
| Epidemiologist (de-identified) | Results, Cases, Clusters, Map, Reports, Extracts | Analysis and reports without names; can generate extracts |
| Administrator | Configuration, Users, Tenants, Reporters, audit log | Thresholds and settings, accounts and access, new counties and programs |
| User administrator | Users, audit log | Accounts and access only, for example at one county |

## 15. Common questions

**My file shows "held 12". Did I do something wrong?**
No. Held rows are MERIT protecting the registry. Open the held link on the batch card, fix each cause once, and release.

**A whole file is held as unknown sender.**
The laboratory's CLIA number is not in the reporter registry. Add the laboratory on **Reporters**, then release the rows.

**I entered a person and then found they already existed.**
Stop adding information to the new record and tell your supervisor, who arranges for the two records to be combined. A combined record is marked read-only with a link to the surviving record, so no history is lost.

**Can I change a result a laboratory reported?**
No. Reported values are kept as sent. Corrections come from the laboratory as a corrected result, or as a correction beside a held row before it is released.

**The Extracts card says Ready but shows findings. Can I send it?**
Findings are warnings. Fix what you can and request the file again; your program decides whether to send a file that still has findings.

**I marked an extract submitted by mistake.**
The file is unchanged and can still be downloaded. Ask your administrator if the status needs correcting.

**A menu item a colleague has is missing for me.**
Your role or profile does not include it. Check the profile in the header, then ask your administrator.

MERIT user guide: daily, weekly and quarterly work. Screens are from the demonstration site; people and data are made up.